AYNA MUSAYEVA

INVESTIGATIVE RESEARCHER | SOFTWARE DEVELOPER

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PROFESSIONAL SUMMARY

Investigative researcher and software developer specializing in corporate intelligence, financial crime research, OSINT, and cross-border due diligence. Develops Python and SQL applications for entity resolution, transaction and graph analytics, evidence verification, and investigative case management.

TECHNICAL & INVESTIGATIVE SKILLS

Investigations: Corporate ownership research, OSINT, due diligence, securities disclosures, regulatory analysis, financial intelligence, evidence reconciliation

Technology: Python, SQL, MySQL, DuckDB, Streamlit, Pandas, graph analytics, Git, Pytest, RAG, FAISS, data validation

PROFESSIONAL EXPERIENCE

Financial & Regulatory Investigative Researcher | Capital, Ownership, and Opacity | Nov 2024–Present

- Investigate cross-border corporate ownership, financial relationships, market structures, and regulatory oversight using filings, securities disclosures, legal materials, and multilingual open sources.
- Reconstruct entity relationships and assess evidence consistency, distinguishing documented facts from inference and unsupported claims.
- Produce research findings and structured analyses on ownership transparency, financing arrangements, and disclosure gaps.

AI Research Infrastructure Developer | Nimillux | Jan 2025–Present

- Developed a local Python research platform combining hybrid retrieval, source-grounded RAG, evidence provenance, and citation validation for heterogeneous document collections.
- Implemented verification controls, confidence assessments, hallucination checks, and regression tests to assess claim support and system limitations.

Marketing Specialist | APOGEE Media | Aug 2023–Apr 2025

- Developed automated marketing campaigns and customer journeys using Salesforce Marketing Cloud, Journey Builder, and Automation Studio.
- Applied SQL, audience segmentation, and Email Studio to support targeted campaigns and CRM workflows.
- Analyzed campaign performance, website traffic, and engagement metrics to optimize digital marketing strategies.

Editor | Musée Magazine | 2018–2019

- Conducted research and fact-checking for print and digital publications.
- Evaluated sources and synthesized findings for editorial features.
- Coordinated open calls, contributor recruitment, and editorial workflows.

SELECTED TECHNICAL PROJECTS

- **CRIE — Corporate Relationship Intelligence Engine:** Built a Python, MySQL, and Streamlit system for GLEIF relationship analysis, entity resolution, evidence integrity, and versioned investigative reports.
- **TRIS — Transaction Risk Intelligence System:** Processed 6.36 million synthetic transactions with explainable alerts, account-network analysis, case management, and held-out fraud evaluation.
- **ATIS — Account Takeover Intelligence System:** Analyzed 31.27 million synthesized login events using authentication anomaly rules, account/IP relationships, evidence tracing, and case audits.
- **BTGI — Bitcoin Transaction Graph Investigator:** Explored 203,769 anonymized Bitcoin transaction nodes and 234,355 directed edges with bounded graph queries and traceable structural alerts.

EDUCATION & CERTIFICATIONS

Harvard — Certificate in Data Science and AI Principles, 2025–2026

B.A., Business Administration — Western University, 2012–2016

Certificate, Documentary Practices & Visual Journalism — International Center of Photography, 2020–2021

Languages: English, Russian, Turkish